SUPPORTING ADULT GUIDANCE

Inequalities as regions on a graph

Inequalities · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This final lesson of the unit extends solution sets from the number line to the coordinate plane. Learners see $x > 2$ swept across every height to fill a region, move a point and watch each rule tick or cross, test whole-number points until the boundary appears, and build regions from up to four rules. It stays with vertical and horizontal boundaries plus $x + y = c$; sloping lines such as $y = 2x + 1$ belong to the Intermediate pathway.

Driving Question: *Where do all the allowed points live?*

The mathematics, in plain terms

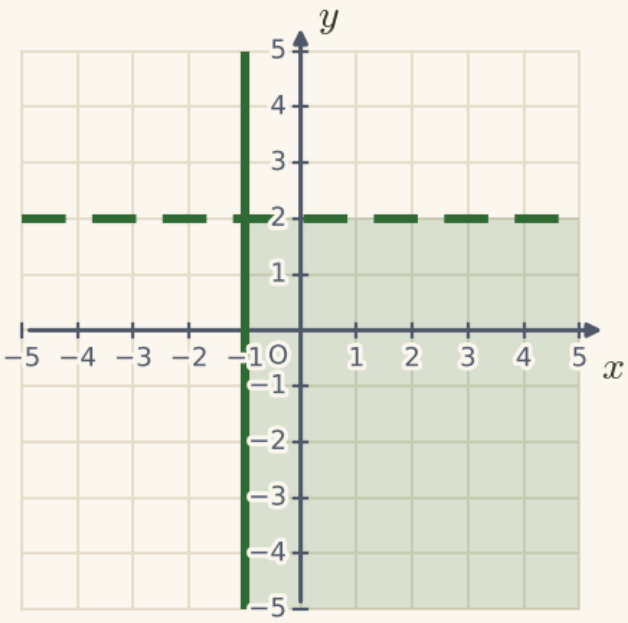
A linear inequality in x and y defines a half-plane bounded by the line obtained by replacing the inequality with equality. The boundary is drawn dashed for a strict inequality (excluded) and solid for a non-strict one (included). The side is found by testing a point not on the line. A system of inequalities defines the intersection of half-planes, and whole-number (integer) points in the region can be listed by systematic checking.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A region from two rules

Show the region where $x \geq -1$ and $y < 2$.

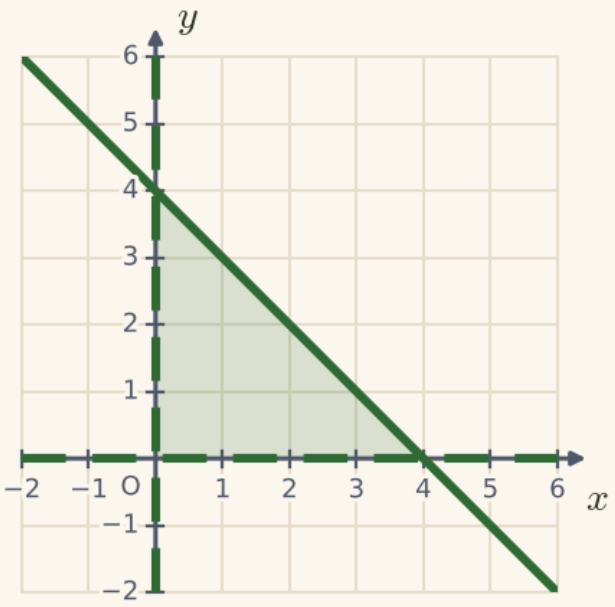


1. $x \geq -1$: draw the vertical line $x = -1$, solid ($\geq$ includes it).	region to the right
2. $y < 2$: draw the horizontal line $y = 2$, dashed ($<$ leaves it out).	region below
3. Shade where both are true.	the overlap

The region right of the solid line $x = -1$ and below the dashed line $y = 2$.

Example 2 — Counting whole-number points

How many whole-number points satisfy $x > 0$, $y > 0$ and $x + y \leq 4$?



1. $x > 0$ and $y > 0$: both coordinates at least 1.	axes left out
2. $x + y \leq 4$: the total is at most 4, so points on $x + y = 4$ count.	solid line
3. List them row by row.	(1, 1) (2, 1) (3, 1) (1, 2) (2, 2) (1, 3)

6 whole-number points.

Points before regions

In Allowed points, learners test individual points before the region is revealed, so the boundary is something they find rather than something they are shown. Let them test freely; the lesson insists on at least six points and some of each kind. When they choose dashed or solid, ask them to test a point on the line — the same check as open or filled circles in Lesson 3.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Sweep a number-line ray up and down the grid until it fills a region.	~6 min

Movement	Drag a point and watch three rules tick or cross; meet four challenges.	~8 min
Reflection	Allowed points: test points, find the boundary, choose dashed or solid, count points.	~16 min
Creativity	The region builder: trap given sets of points with up to four rules.	~10 min
Rest	A pause to close the unit: one idea seen four ways.	~2 min
Nutrition	Intellectual: noticing when an old idea grows into a new space.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Drawing $x > 2$ as a horizontal line (confusing $x = 2$ with $y = 2$).
Ask for three points with $x = 2$, plot them, and ask which way the line through them runs.

Dashed and solid lines the wrong way round.
Test a point on the line: for $x > 2$, is (2, 0) allowed? Link to open and filled circles.

Shading the wrong side.
Test (0, 0) (or any point not on the line) in the inequality, and shade the side it is on if it is true.

Including points on a dashed boundary when listing whole-number points.
Check each listed point in every rule, especially those on a line.

Thinking a point needs to obey only one of several rules.
Use the wandering point: 'How many ticks do you need for the point to be allowed?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: plotting coordinates in all four quadrants, and open and filled circles from Lesson 3. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner shades the region for $y \leq 1$ above the line. The aim is a question that lets them find the slip themselves.

- "Pick a point in your shaded region, say (0, 3). Is $3 \leq 1$ true?"

- “Now try (0, 0). Is that one allowed?”
- “Which side of the line is (0, 0) on?”
- “Is the line itself included? How did you decide?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why not sloping lines like $y = 2x + 1$?

Those belong to the Intermediate pathway. This lesson keeps to vertical and horizontal boundaries and $x + y = c$, so that the focus stays on regions, boundaries and testing points.

Should learners shade the region they want or the region they do not want?

This lesson shades the wanted region. Some exam questions use the opposite convention; reading the question carefully is the habit to build.

Why does the region builder allow four rules?

A rectangle needs four boundaries. Three are enough for a triangle. Choosing how many rules to use is part of the design.

What comes next?

This is the last lesson of the unit. A short end-of-unit assessment follows.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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